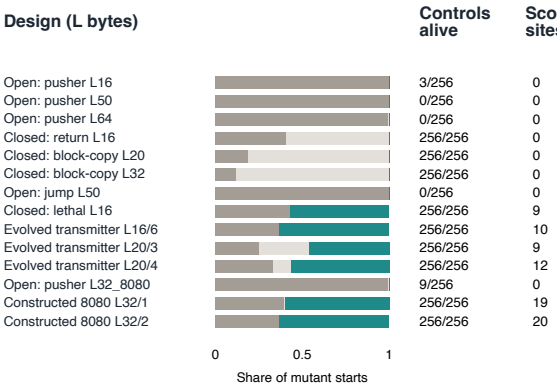
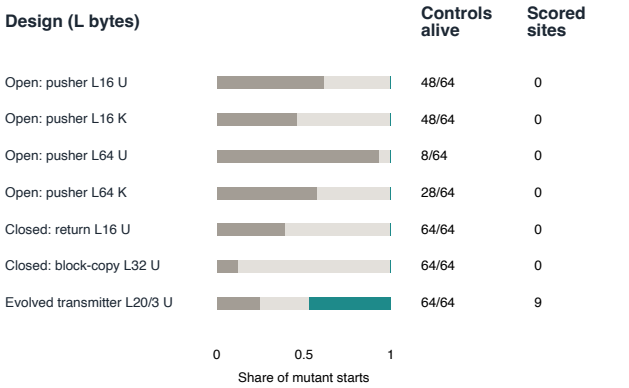
a All historical serial designs: transfer 8



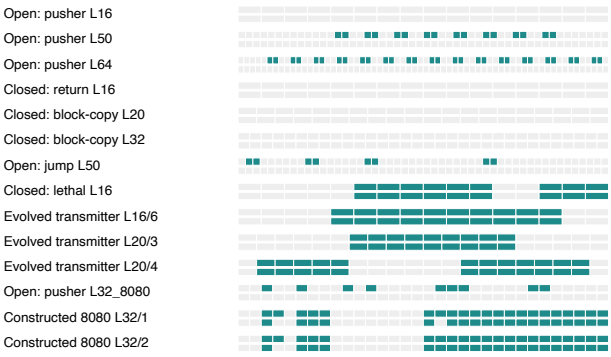
Dark grey lost; light grey alive <50%; teal alive ≥50% carrying
Carriage = members containing mutant window / population members
Site: mutant-survivor carriage − WT-survivor carriage ≥0.5 at ≥half values

b Branching generation-8 outcomes: U / K



Same partition/site key as a; controls = unmutated populations
U uniform-byte; K archived soup partners; all archived pairs shown
32 values/site ×4 populations/value; 64 controls/design

c Complete serial site maps: 1 / 8



Upper track: transfer 1; lower: transfer 8; teal = scored site
Site criterion as a; each row spans byte 0 to L−1 (L in label)

d All partner-environment carriage tests



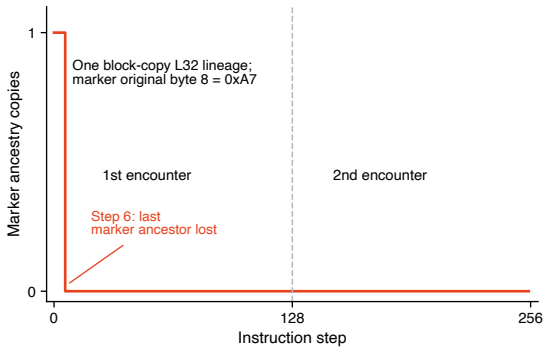
Serial: one descendant/transfer; g8 carrier contains mutant window
Starts/design = L×255×16; L bytes in labels; 256 WT controls
Benign zero executes / lethal halts; open soup at 5,000 updates

e Identifiable alternatives and controls

Design	Ident. / byte alternatives	WT starts
Open: pusher L16	4080/4080	256
Open: pusher L50	1600/1600	256
Open: pusher L64	2048/2048	256
Closed: return L16	4080/4080	256
Closed: block-copy L20	5100/5100	256
Closed: block-copy L32	8160/8160	256
Open: jump L50	1600/1600	256
Closed: lethal L16	4080/4080	256
Evolved transmitter L16/6	4080/4080	256
Evolved transmitter L20/3	5100/5100	256
Evolved transmitter L20/4	5100/5100	256
Open: pusher L32_8080	8160/8160	256
Constructed 8080 L32/1	8160/8160	256
Constructed 8080 L32/2	8160/8160	256

Identifiable = marker window absent from wild type (WT)
Alternative types = L byte positions × tested values
16 starts/value; pusher50/64 & jump50: 32 values, others 255

f Ordered ancestry: two full encounters



Counted unit: bytes with byte-8 ancestry; all-zero byte partners
Step 6 reads address 4 (ancestry byte 0), writes 0x04 over 8
At 128: offspring becomes parent; processor reset